

# Co-MLOps Platform INTRODUCTION



 **Co-MLOps**

Collaborative Data and AI Platform

For AD / ADAS

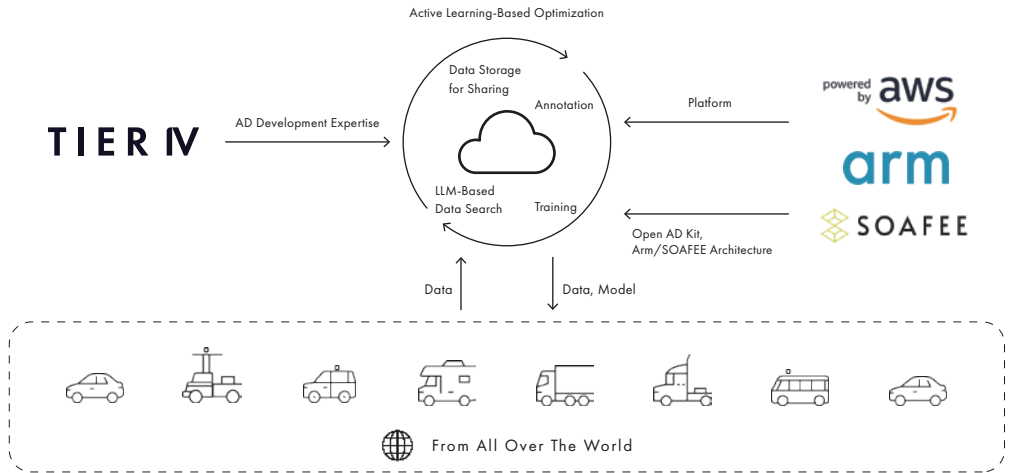
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# About Co-MLOps Platform

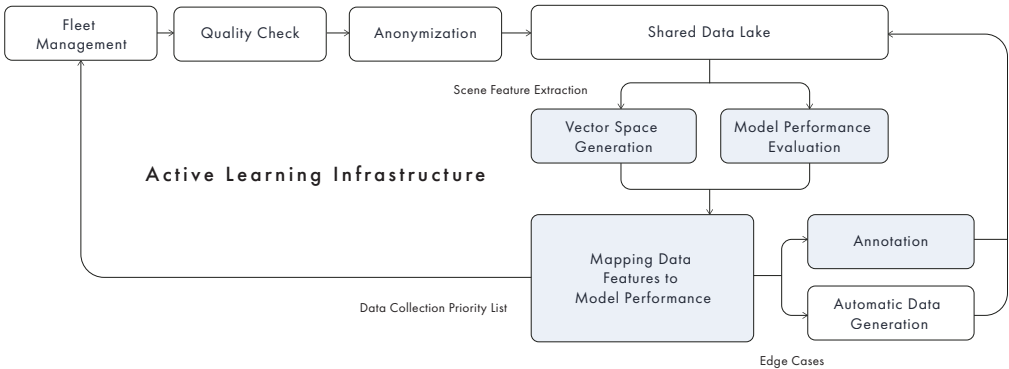


The Co-MLOps Platform has been designed to scale the development of autonomous driving AI.

Developed by TIER IV, the platform makes it possible to accelerate AI development using large-scale shared data collected from around the world.

## Workflow

Sensor data such as camera images and LiDAR point clouds collected from various regions around the world can be shared through the Co-MLOps Platform. By utilizing the provided MLOps features and edge AI reference models, companies can enhance their autonomous driving AI development.



# Large-scale Data Collection

## Featuring reference sensor suite and anonymization

### Data Collection



To support the rapid development and optimization of autonomous driving systems, we offer comprehensive services including advanced data collection, cloud management, and image anonymization for privacy protection.

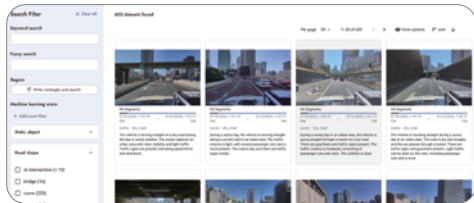
Data collection vehicles are equipped with time-synchronized and calibrated sensors such as cameras and LiDAR, enabling high-precision recording of road conditions and surrounding environments.



#### Automotive Cameras

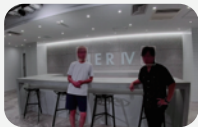
The Co-MLOps Platform utilizes autonomous driving cameras developed by TIER IV.

### Cloud Data Management



The collected data is securely and efficiently managed in a cloud-based system, allowing for data retrieval services and both automated and manual annotation management.

### Anonymization Technology



Face Blurring



Face Swapping

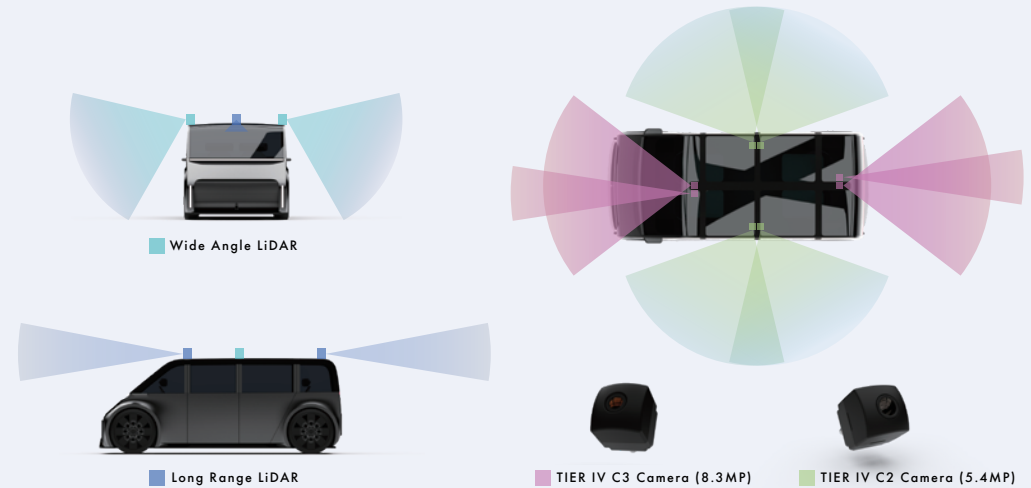
Privacy is safeguarded by anonymizing license plates and human faces.

Vast amounts of data are collected from vehicles during autonomous driving. Protecting privacy is crucial.

On the cloud side, data anonymization ensures secure data management, while on the edge side, real-time data processing and anonymization at the point of collection reduce privacy risks.

# Data Recording System (DRS)

A robust suite of sensors and computing systems for advanced data collection.



#### Camera

Supports 360-degree long-range coverage around the vehicle using TIER IV's high-resolution HDR cameras.

#### LiDAR

Captures high-density point clouds in 360 degrees. Each sensor is synchronized and can be used as a multimodal ground truth.

#### Radar

Enables robust perception development, unaffected by weather changes.

#### GNSS / IMU

Provides vehicle position and acceleration data, ensuring continuous accurate location information.

#### Vehicle CAN

Delivers information on vehicle operations.

#### Data transfer

Supports easy upload to a cloud-managed data platform.

DRS

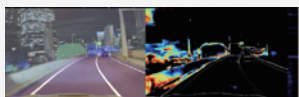
# Active Learning Framework

## Uncertainty Estimation



Using the collected dataset, uncertainty and reliability are quantified. Specifically, the model's confidence in its predictions is assessed, and areas with high uncertainty are identified to improve the system's accuracy and reliability.

### Active Learning Features for AI Training



Uncertainty Estimation



Parallel Coordinates

Active learning features enable efficient data selection for training, even with a small dataset. Labeling and training in areas with high uncertainty are prioritized, among other factors.

## Reference AI

### Full Surround Perception for AD and ADAS



The reference AI model is a lightweight model that supports common recognition functions used in AD/ADAS applications. It serves as a foundational model upon which developers can build their own algorithms, facilitating efficient development of advanced autonomous driving systems.

# Annotation Samples

The platform supports three types of annotations.

## Panoptic Segmentation



Panoptic Segmentation is a technique that assigns pixel-level labels to every object in an image. It distinguishes not only the shapes of objects but also the background and different instances of the same type of object. This enables a more detailed and complex understanding of scenes compared to other annotation methods, allowing it to handle a wider range of tasks.

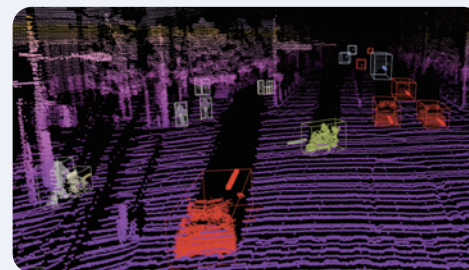
## Traffic Light



The platform supports the development of various traffic signal recognition systems, both domestically and internationally, with traffic signal regions enclosed in bounding boxes and each signal light labeled with attributes such as color, type, and arrow direction.

- Traffic signal detection
- Traffic signal state recognition  
(e.g., signal color and lighting status)
- Traffic signal type identification  
(e.g., pedestrian signals or right-turn indicators)

## 3D Bounding Box



Based on the 3D point cloud data collected by LiDAR, a bounding box is defined in the three-dimensional space to encompass the position and shape of an object. This box is used for object detection and tracking.

- Object detection and classification
- Object localization
- Size and orientation information

# Partnerships

## Scalable & Secure

### Hitachi Astemo



Co-MLOps Platform is driving the development of a large-scale data-sharing infrastructure to accelerate data utilization and management in autonomous driving. It has already produced numerous success stories and collaborates with many industry leaders.

Since participating in the initial Co-MLOps pilot program, which began in 2023, involving data collection across eight global regions, we have been collaboratively advancing the development of key functionalities of the platform as a lead partner since 2024.

Furthermore, we are jointly pushing forward the development in the fields of autonomous driving platform and Software-Defined Vehicles (SDV).

### Nihon Kotsu



Since July 2024, Nihon Kotsu and TIER IV have been collaborating to promote the development of a large-scale shared data infrastructure.

By utilizing vehicles equipped with TIER IV's Data Recording System (DRS), they are jointly building extensive and diverse datasets, contributing to the development of safe autonomous driving technology.

### The Autware Foundation (AWF)



The Autware Foundation was established in 2018. Autware is one of the world's leading open-source projects for autonomous driving, and TIER IV actively contributes to its development and community management.

## Get on board

By joining the Co-MLOps Platform, you can leverage a scalable and secure data-sharing infrastructure powered by cutting-edge AI technology.



The platform features innovative technologies, including an active learning infrastructure that enables efficient data labeling and large-scale AI training. Let's share data together and accelerate business growth and technological innovation.

### Benefits:

- Strengthened collaboration through data sharing
- Cutting-edge AI technology and active learning infrastructure
- Scalable data infrastructure
- Top-level security and data protection
- Technical support and customization
- Joint innovation

### Contact Us

**TIER IV Tokyo Office**  
Jacom Building, 1-12-10 Kitashinagawa,  
Shinagawa Ward, Tokyo 140-0001, Japan

[co-mlops@tier4.jp](mailto:co-mlops@tier4.jp)

